Literature Status Note

Problems A–C from arXiv:1012.4514

Run `fa_banach_001` — lane 5

27 August 2026

LITERATURE ALREADY ANSWERED — PROBLEMS A, B, AND C

Source questions

Levy and Shalit's survey *Dilation theory in finite dimensions* asks four labeled questions [1]. This note identifies exact later answers to the first three.

- **Problem A** (source PDF page 7): can a commuting tuple of contractive 3×3 matrices violate
$$\|p(T_1, \dots, T_d)\| \leq \sup_{z \in \mathbb{T}^d} |p(z)|?$$
- **Problem B** (source PDF page 8): are Brehmer's positivity inequalities equivalent to the existence, for every N, of a regular unitary N-dilation on a finite-dimensional space?
- **Problem C** (source PDF page 9): does every pair of commuting contractions on a finite-dimensional space have a finite-dimensional unitary N-dilation for each fixed N?

Exact literature answers

Problem A: no counterexample exists. Knese's Theorem 1.2, on supporting PDF page 2, states that the von Neumann inequality holds for every d-tuple of commuting contractive 3×3 matrices [2]. This is precisely the universal negation of the existence question in Problem A. Knese attributes the decisive three-point Pick interpolation input to Kosiński and supplies the operator-theoretic reductions.

Problem B: yes. McCarthy and Shalit's Theorem 1.7 and Corollary 1.8, on supporting PDF page 4, prove that a finite-dimensional commuting tuple has finite-dimensional regular unitary N-dilations for every N exactly when it has a regular unitary dilation; Brehmer's inequalities characterize the latter [3]. Their corollary is word-for-word the equivalence requested in Problem B.

Problem C: yes. McCarthy and Shalit's Theorem 1.2, stated on supporting PDF page 2, proves more generally that whenever a commuting tuple on a finite-dimensional space has a unitary dilation, it has a finite-dimensional unitary N-dilation for every fixed N [3]. Apply Ando's dilation theorem to a pair of commuting contractions.

Proof intuition for the finite-dimensional reduction

The common mechanism behind Problems B and C is finite moment compression. A commuting unitary dilation yields an operator-valued measure on the torus. Only finitely many moments are needed at degree N. A matrix-valued Carathéodory argument replaces that measure by a finite atomic positive operator-valued measure with the same moments, and finite-dimensional Naimark dilation turns its atoms into commuting unitaries. The regular version retains both positive and negative Laurent moments. This is the exact finite-dimensional bridge that the source paper sought.

Scope boundary

Problem D asks for a "commutant lifting theorem" for unitary N-dilations, without fixing a unique formulation. McCarthy and Shalit explicitly say that the appropriate analogue is unclear and show that one natural fixed-dilation version fails already for a particular Halmos 1-dilation [3, Sec. 2.3]. That observation does not settle every plausible existential version. We therefore record Problems A–C as answered and make no durable status claim for Problem D.

Review recommendation

Classify the queued Problem-A signal as answered by Knese. Record Problems B and C as additional exact answers from McCarthy–Shalit. Do not spend a new proof attempt on this paper unless Problem D is first reformulated precisely.

References

[1] E. Levy and O. M. Shalit, *Dilation theory in finite dimensions: the possible, the impossible and the unknown*, arXiv:1012.4514; Rocky Mountain J. Math. 44 (2014), 203–221.

[2] G. Knese, *The von Neumann inequality for* 3×3 *matrices*, arXiv:1508.06488; Bull. Lond. Math. Soc. 48 (2016), 53–57.

[3] J. E. McCarthy and O. M. Shalit, *Unitary N-dilations for tuples of commuting matrices*, arXiv:1105.2020; Proc. Amer. Math. Soc. 141 (2013), 563–571.